

MENG WEI

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Overview

- Applied AI researcher handling real-world multimodal data (biomedical signals, clinical records, medical imaging, surgical video) with a strong mathematical (statistical modeling, optimization) foundation.
- Experienced with computer vision, efficient pre-training, multimodal modelling, NLP, VLM, graph neural networks, semi-/ weakly supervised learning, reinforcement learning, and multi-agent systems.
- Deep learning practitioner with end-to-end experience in model design, evaluation, and implementation with PyTorch (and some TensorFlow and JAX), Lightning, HuggingFace, Transformers, and HF Datasets.
- Strong research engineer experienced with project management, git, experiment logging, distributed training (FSDP, DDP), programming (C, C++, Verilog), and data pipelines (BASH, SCALA).

Work Experience

Research Fellow | UHN & Vector Institute, Toronto, Canada

Jan 2026 – Present

- Building a privacy-preserving evolvable multi-modal biomedical multi-agent system supporting multi-task learning for both clinical and research use, with continual fine-tuning via reinforcement learning.
- Leading end-to-end development of an agent-based operating system for biomedical AI, spanning training, optimization, and deployment, with optimized memory management.

Machine Learning Engineer Intern | Viridien (CGG), London, UK

Jun 2021 – Feb 2022

- Applied generative models and deep learning to 3D seismic data for synthetic data generation and volumetric segmentation in an energy industry setting.
- Delivered production-ready AI solutions for the energy sector in direct client collaboration.

Summer Research Intern | AI in Medical Image Lab, UCLA, USA

Jul 2019 – Sep 2019

- Proposed a deep learning framework for hemorrhagic stroke detection using symmetry constraints and adversarial training to improve lesion segmentation for clinical diagnosis.

Education

PhD AI for Medical Imaging | King's College London

Oct 2021 – Nov 2025

UK Center of Doctoral Training - Awarded full 4-year funding & 3-month post-doc transitional funding for international doctoral study by the UK Engineering and Physical Sciences Research Council (EPSRC)

- Conducted fundamental and applied research in AI for medical imaging, focusing on developing new ML models and AI techniques for challenging real-world clinical problems.
- Developed segmentation and hierarchical instance segmentation frameworks using semi-supervised and weakly-supervised strategies for data-efficient model optimization.
- Designed novel lightweight adapters for efficient foundation model adaptation to medical imaging.
- Constructed VLMs for surgical image and video understanding, incorporating domain-specific strategies (motion-centric spatiotemporal reasoning, reinforcement module) with multi-institutional dataset curation.

MRes Artificial Intelligence & Machine Learning | Imperial College London

Oct 2020 – Oct 2021

Medical Robotics and Image-Guided Intervention pathway: Distinction (Outstanding Prize)

- Individual Project: Improved small lesion detection in CT via meta-learning leveraging large-lesion data.

BEng Electrical Engineering | University of Liverpool

Sep 2017 – Jul 2020

Telecommunication Engineering pathway: First-class honours

- Meritorious winner of The Mathematical Contest in Modeling (MCM), 2018
- Vice president of the Women in Engineering Society
- Final-year project: CNN-GCN aggregation-enabled boundary regression model for image segmentation.

Selected Publications

- Wei, M., et al. Referring Surgical Instrument Segmentation via Motion. *AAAI*. 2026.
- Wei, M., et al. Surgical Instances and Part Semantics for Weakly-supervised Part-aware Instance Segmentation. *IEEE Transactions on Medical Imaging*, major revision. 2025
- Wei, M., et al. Semi-supervised surgical instrument segmentation. *Nature Scientific Reports*. 2025
- Wei, M., et al. Enhancing surgical instrument segmentation with adapters. *IJCARS*. 2024.
- Zhou, Z., Wei, M., et al. Text Promptable Surgical Instrument Segmentation with VLMs. *NeurIPS*. 2023.
- Meng, Y., Wei, M., et al. Regression of Instance Boundary by Aggregated CNN and GCN. *ECCV*. 2020.